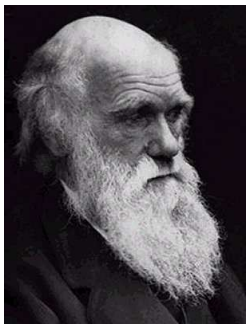
Evolution

All the different life forms we see, started to evolve how many years ago?

1 billion ☐ 2 billion ☐ 3 billion ☐

What did all life start out as? _____



Charles Darwin was an English naturalist. What was the name of his famous book? Origin of _____

This book describes how animals change over time due to N_____ S_____

Evolution via natural selection is the outcome of four points. What are they?

1. _____
2. _____
3. _____
4. _____

Natural selection is basically survival of the fittest. Explain?

_____

_____

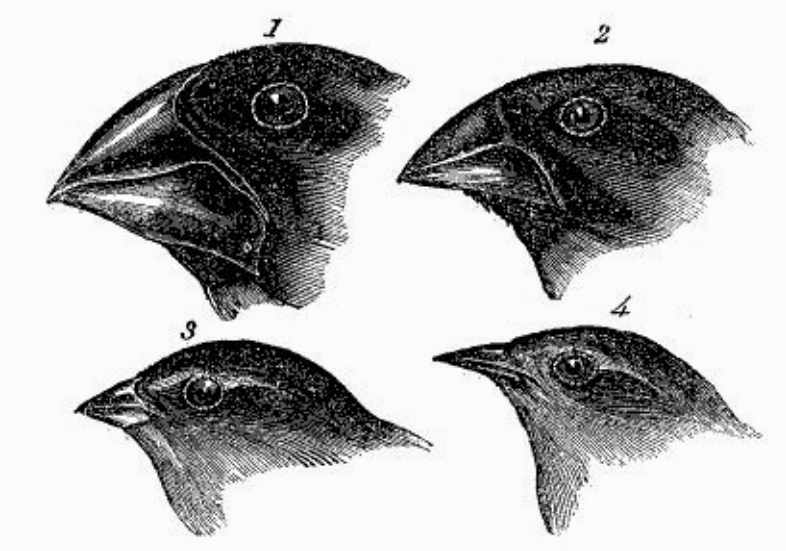
It was on his voyage on the HMS Beagle in 1831 that Charles Darwin visited the G_____ Islands.



Darwin's finches

There are 13 species of finch on the Galapagos island.
Their beaks differ in two characteristics, what are they?

_____ _____



Why have these finches evolved different beak size and shapes?

_____

_____

How do you think these finches evolved different beak size and shapes?

_____

_____

_____

Match the beak to the type of food the finch eats.









Mainly insects
Buds and fruit
Cactus seeds and parts
Mainly seeds

Theories of Evolution

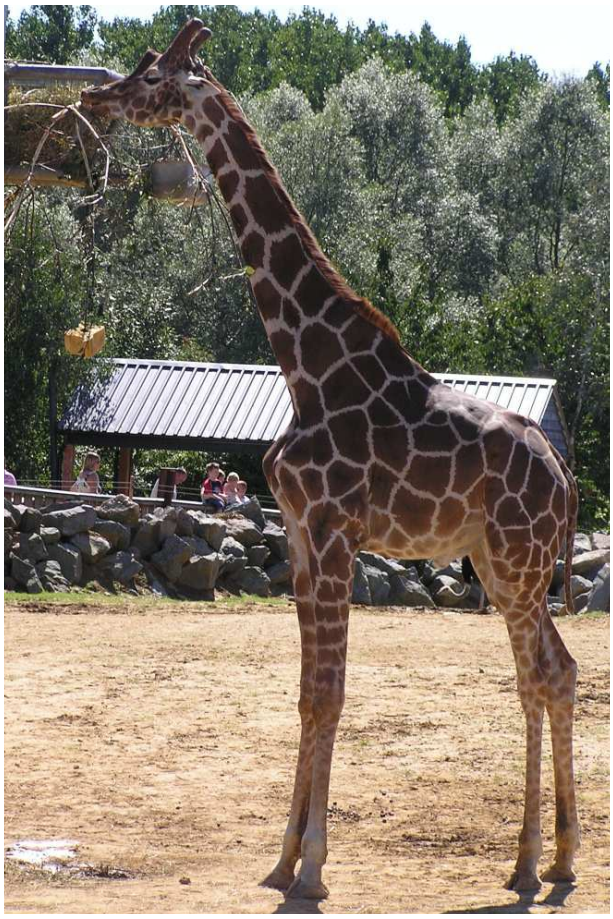
Darwin and Lamarck proposed different theories of evolution, while Darwin explained evolution by natural selection Lamarck's alternative theory included two main ideas:

- The law of use or misuse
- The law of inheritance of acquired characteristics

Fill in the boxes below to show how Darwin and Lamarck would explain how giraffes evolved to have long necks.

Darwin

Lamarck



Can you identify and explain a third alternative explanation of evolution?

_____

_____

_____

Evidence for Evolution: Fossils

What is a fossil? _____



How are fossils made?

_____

_____

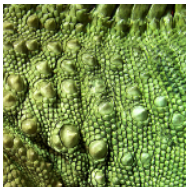
_____

Is the fossil record complete? Yes / No

Group these items to what would fossilise and what would not fossilise:

Fossilise	Wouldn't fossilise





How do rocks and soil provide evidence for evolution?

_____

_____

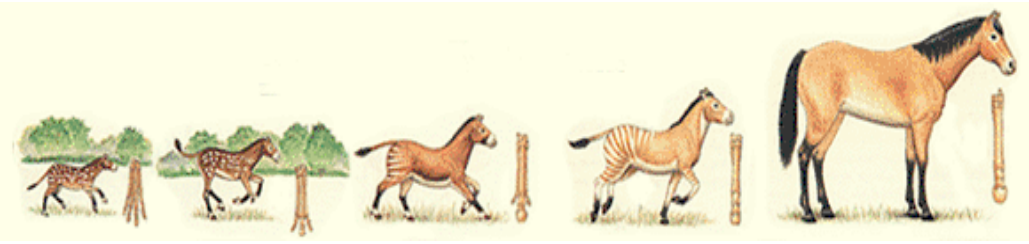
The fossil record for horse evolution is complete. Explain how horses evolved using the key words below.

habitat *running* *see predators* *evolution* *natural selection*

_____

_____

_____



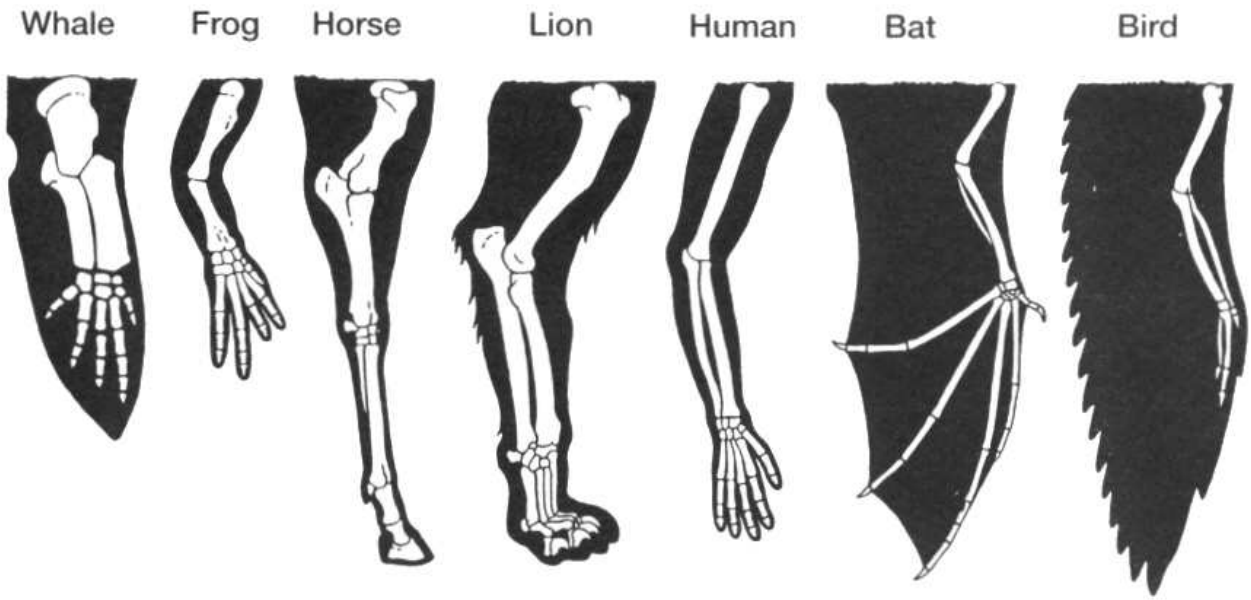
Evidence for Evolution: Anatomy

What is a vertebrate? _____

Why do you think all vertebrates have similar bone structures in the hand?

_____

_____



How do bats and birds differ in their bone structure?

_____

_____

How do you think the whale evolved a fin like structure?

_____

_____

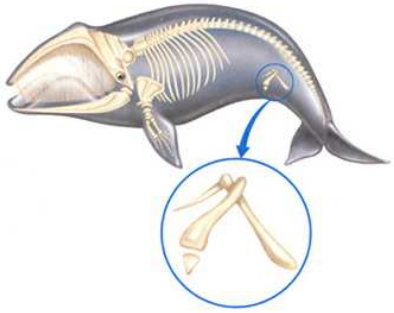
Whales evolved from a land animal. How do we know this?

_____

_____

What name do we give to bones or organs which remain as evidence for evolution?

V_____ O_____



Do they serve any purpose now? Yes / No

Evidence for Evolution: The Peppered Moth

Before the British industrial revolution, the peppered moth was pale. The birch trees in its habitat were also pale. Sometimes, a mutation happened and a dark peppered moth would appear.



How did the survival rates of the pale and dark peppered moth differ before the British industrial revolution?

_____

_____

_____

_____

Due to the amount of airborne pollution during the British industrial revolution; most of the birch trees turn black due to the soot.



How did the survival rates of the pale and dark peppered moth differ after the British industrial revolution?

_____

_____

_____

_____

Mutations are not always good.



A white tiger is the result of a mutant gene.

Would a white tiger survive in the wild?

Yes / No

Explain using the keywords below:

camouflage *natural selection* *starvation* *reproduce*

_____

_____

_____

_____

Evidence for Evolution: Antibiotic resistant bacteria

How do you think some bacteria evolve to become resistant to antibiotics? _____

_____

_____

This can happen very quickly. Why?

_____

_____



Why is this a problem?

_____

_____

Selection

Evolution tends to occur via what type of selection?

Natural or Artificial? _____

What is the difference between natural and artificial selection?

_____

_____

_____

What type of selection do farmers use to ensure their hens lay big eggs or their cows produce lots of milk? _____

Give reasons as to how artificial selection is a good thing and bad.

Good	Bad

Human Evolution

What group of animals are humans thought to have evolved from? _____

What is the difference between a monkey and an ape?

Monkey	Ape

Within the great apes, what are humans more closely related to?

Orangutans ☐ Gorilla ☐ Chimpanzee ☐

It has been shown that throughout human evolution, brain size has significantly increased.

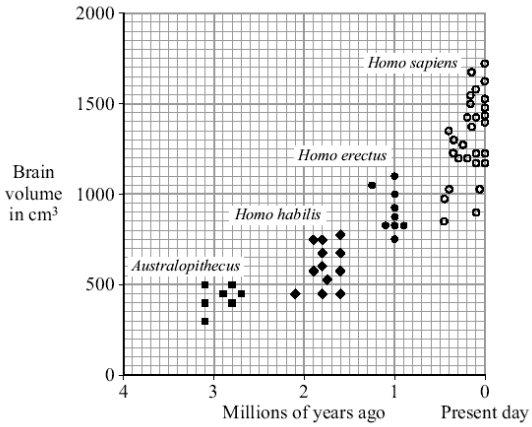
Why do you think brain size increased?

_____

_____

_____

_____



What does a larger brain size mean in terms of behaviour and problem solving? _____

_____

_____



Why do you think we evolved to walk upright on two legs? _____

_____

_____

_____